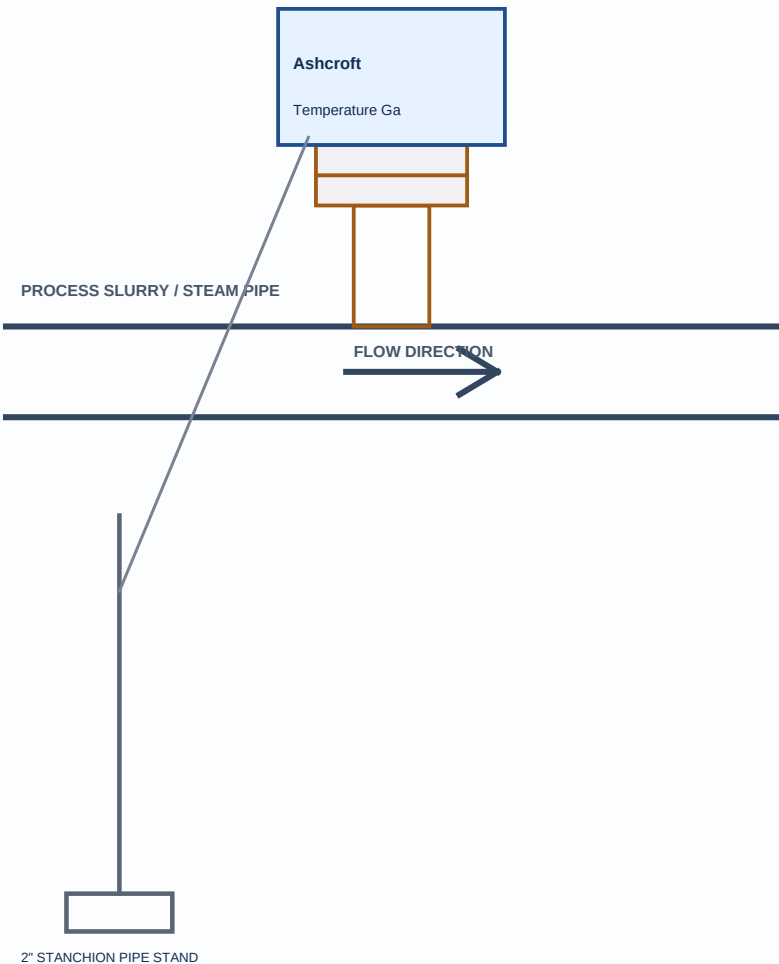


TEMPLATE TYP-18: TEMPERATURE GAUGE

TYPICAL TAGS: TI-40101 / TI-60201

SERVICE: Direct Pipe & Tank Slurry Temperature Indicator | PLC I/O: None (Local Operator Readout)

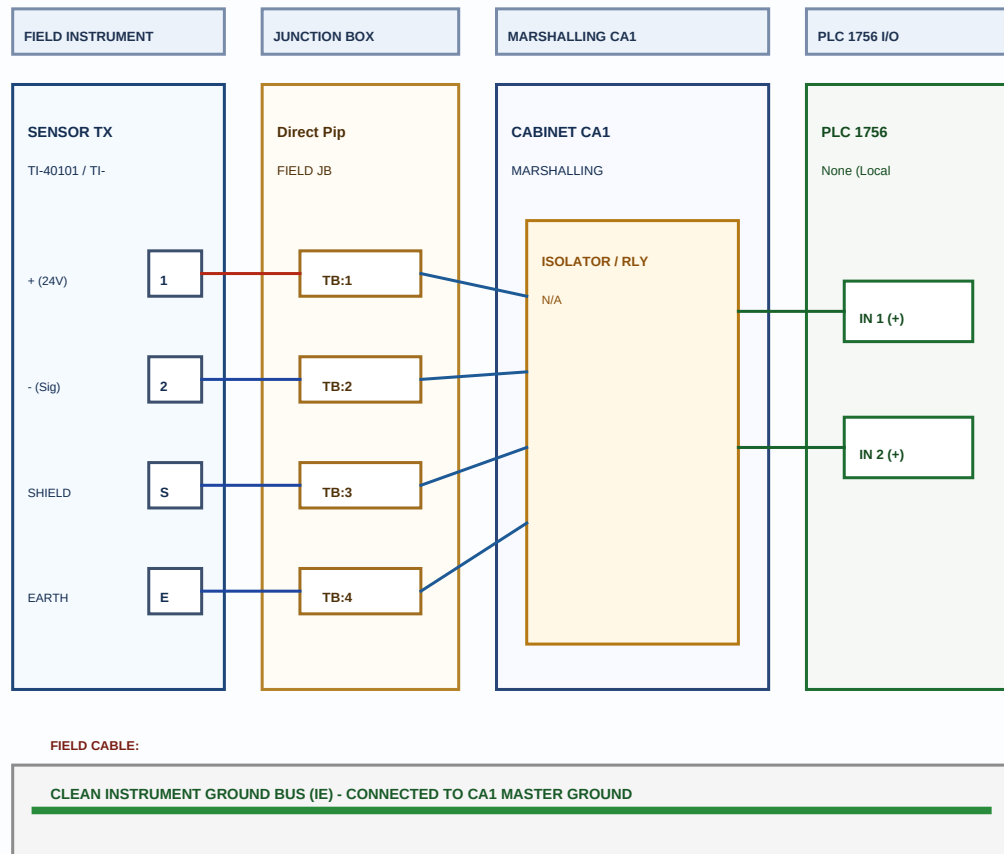
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	1/2" NPT Male Thread into Ba..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	IP65 Hermetically Sealed Sta..
Hazardous Class:	Non-Electrical Apparatus (St..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



FIELD CABLE:

CLEAN INSTRUMENT GROUND BUS (IE) - CONNECTED TO CA1 MASTER GROUND

ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	Visual Scale 0-150°C (32-300°F)
Power Supply:	None (Bimetal Coil Driven)
Field Cable Type:	N/A (Local Dial Readout)
Destination JB:	Direct Piping Well Mount
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	N/A
PLC Module:	None (Local Operator Readout)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ASHCROFT

MODEL: Model FI (100mm Dial Everyangle Bimetal) + Thermowell

SERVICE: Direct Pipe & Tank Slurry Temperature Indicator

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Bimetal Thermometer Model FI (Ashcroft 100mm Dial Everyangle S		1 EA
2	Barstock Thermowell SS316 (U=1 1/2" NPT x 1" NPT Process Thread		1 EA
3	Weldolet / Thredolet 1" NPT ASTM A182 F316 Pipe Fitting		1 EA
4	Thermal Conductive Paste High Temperature Aluminum Oxide		1 TUBE
5	Safety Wire & Tag Plate SS316 Laser Engraved Tag Plate		1 EA

ENGINEERING INSTALLATION NOTES

- Thermowell tip must be immersed into center third of process pipe for representative measurement.
- Apply thermal conductive paste inside thermowell bore before inserting gauge stem to improve heat transfer.
- Thermowell allows removal of thermometer for calibration without process shutdown.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONBIMETAL TEMPERATURE GAUGE & THERMOWELL HOOK-UP DETAIL
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-018
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
20 / 25